

Athena AI Platform

Complete Sidebar Features & Functional Specification Guide

This official reference document details the purpose, technical architecture, business value, and primary capabilities of every operational module accessible from the **Athena AI Sidebar Navigation Menu**.

Sidebar Icon & Feature	Primary Function & Description	Target Audience
■ Dashboard	Real-time telemetry, live WebSockets metrics, memory & department token usage	DevOps, Admins, CTOs
■ Chat Interface	RAG document Q&A;, context memory, streaming answers, document search	Analysts, Knowledge Workers
■ ML Classifier	Automated machine learning transaction classification & confidence scoring	Finance, Accountants, Auditors
■ Workflow Builder	Visual drag-and-drop orchestration of multi-step AI & code execution pipelines	Automation Engineers, Analysts
■ API Hub	Enterprise connectors, external REST tools, database webhooks, and secret key vault	Integrators, IT Admins
■ Security	Multi-tenant isolation, RBAC department clearance, JWT auth, security logs	CISOs, Compliance Officers
■ Performance	GZip compression telemetry, Redis caching hit-ratio, FastAPILimiter rate control	SREs, Infrastructure Engineers
■■ Settings	LLM provider selection (Gemini, OpenAI, Ollama), user profile & theme preferences	All System Users

1. ■ Analytics Dashboard

The **Analytics Dashboard** serves as the central command bridge for monitoring platform performance, operational stability, and multi-tenant resource utilization in real time.

Key Capabilities:

- **Live System Telemetry:** Establishes a persistent WebSocket connection to /api/v1/metrics/live streaming live CPU/RAM utilization, response latency, and active execution counts.
- **Token Usage by Department:** Bar chart analytics tracking LLM token consumption across Finance, Legal, HR, Engineering, and Sales for cost allocation.
- **Workflow Execution Rate:** Pie chart visualizing execution stability (e.g. 85% success vs 15% failure rate).
- **Real-Time Latency Time-Series:** Time-series line chart updating every second to monitor API responsiveness.

2. ■ Chat Interface

The **Chat Interface** is a full-screen conversational assistant powered by Retrieval-Augmented Generation (RAG) and multi-tenant vector memory.

Key Capabilities:

- **RAG Document Search:** Allows users to query uploaded enterprise documents (PDFs, DOCX, CSVs) with page citations.
- **Streaming Token Delivery:** Uses Server-Sent Events (SSE) to stream model responses word-by-word with zero delay.
- **Session History Management:** Persists conversation threads with instant creation, selection, and deletion.
- **Multi-Tool Execution:** Automatically invokes python execution, web search, or financial database lookups as required.

3. ■ ML Expense Classifier

The **ML Expense Classifier** provides automated Machine Learning classification of raw transaction descriptions into enterprise financial accounting categories.

Key Capabilities:

- **Automated Category Inference:** Analyzes raw invoice lines (e.g. 'AWS Cloud Compute') and categorizes them into standard accounting buckets.
- **Confidence Scoring:** Outputs model confidence percentages (e.g. 98.4%) to guarantee audit accuracy.
- **Batch Invoice Processing:** Ingests multi-row transaction payloads for high-speed ledger posting.

4. ■ Workflow Builder

The **Workflow Builder** enables users to visually design, schedule, and execute complex multi-step AI automations and asynchronous background tasks.

Key Capabilities:

- **Multi-Step Pipeline Orchestration:** Links LLM prompts, Python code sandboxes, SQL queries, and external APIs.
- **Asynchronous Celery Execution:** Offloads long-running tasks to background Celery workers backed by Redis queues.
- **Execution Status Tracking:** Real-time status indicators for Pending, Running, Success, and Error states.

5. ■ Enterprise API Hub

The **Enterprise API Hub** acts as the central integration portal for managing external SaaS connections, webhooks, and REST tools.

Key Capabilities:

- **Third-Party Connectors:** Pre-built tool integrations for Slack, Salesforce, Google Drive, PostgreSQL, and custom REST APIs.
- **Secret Vault & Keys:** Encrypted storage for API tokens and authentication headers.
- **Webhook Event Listeners:** Listens for external triggers to launch automated AI workflows.

6. ■ Enterprise Security

The **Security Module** handles Role-Based Access Control (RBAC), tenant data isolation, and authentication security.

Key Capabilities:

- **Department RBAC Isolation:** Enforces data boundaries between Finance, Procurement, Admin, and HR departments.
- **Bcrypt & JWT Security:** Hashed passwords and signed JSON Web Tokens for stateless session verification.
- **Security Audit Logging:** Records user logins, document uploads, and access events for compliance.

7. ■ Performance & Cache

The **Performance Dashboard** provides granular monitoring of network optimization, Redis caching hit rates, and rate limiting.

Key Capabilities:

- **GZip Compression:** Middleware metrics for automatic network payload compression on responses > 1KB.
- **Redis Cache Telemetry:** Tracks cache hits for vector embeddings and frequent query responses.
- **FastAPILimiter Rate Control:** Limits request rates per client to prevent API abuse.

8. ■■ System Settings

The **Settings Module** allows users and administrators to customize LLM providers, model parameters, and UI themes.

Key Capabilities:

- **LLM Provider Engine Switching:** Seamlessly switch model backends between Google Gemini, OpenAI GPT-4, and Local Ollama Llama 3.2.

- **User Profile Management:** Update username, email preferences, and password credentials.
- **Theme Customization:** Dark Mode and glassmorphic UI customization controls.